

Exercise 1: (6 Points) Decide for each of the following statements whether they are true or false. *Hint:* You will get points for correctly answered questions and points subtracted for wrong answers. However, you can not achieve negative points in a given question block.

a) Block I (2 Points)

- i) $rk(\mathbf{X}'\mathbf{X}) = p$.
☐ TRUE ☐ FALSE
- ii) The average of the predicted values is equal to the average of the observed response.
☐ TRUE ☐ FALSE
- iii) For the test of significance with $H_0 : \beta_j = 0$, it holds that under H_0 , $t_j^2 \sim F_{1,n-p}$.
☐ TRUE ☐ FALSE
- iv) In Q-Q plots, the empirical quantiles are compared to the quantiles of the theoretical distribution. If the data follows the distribution, the points should closely follow a horizontal line through the origin.
☐ TRUE ☐ FALSE

b) Block II (2 Points)

- i) The relation $y = \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon)$ cannot be analysed within the linear regression framework.
☐ TRUE ☐ FALSE
- ii) $rk(\mathbf{X}'\mathbf{X}) = k$.
☐ TRUE ☐ FALSE
- iii) The ridge estimator adds a constant λ to every component of $\mathbf{X}'\mathbf{X}$ to obtain a regularized and invertible matrix.
☐ TRUE ☐ FALSE
- iv) For a general linear hypothesis with $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, it holds that under H_0 , $F \sim F_{p+1,n-p}$.
☐ TRUE ☐ FALSE

c) Block III (2 Points)

- i) Since the residuals are normally distributed, standardized residuals are also normally distributed.
☐ TRUE ☐ FALSE
- ii) For the test for significance of regression with $H_0 : \beta_1 = \dots = \beta_k = 0$, it holds that under H_0 , $F \sim F_{p-1,n-p}$.
☐ TRUE ☐ FALSE
- iii) Interaction can only be computed between two continuous or between a continuous and a categorical independent variable.
☐ TRUE ☐ FALSE
- iv) $rk(\mathbf{X}'\mathbf{X}) = n$.
☐ TRUE ☐ FALSE

Data set for the following Exercises 2 & 3

Within the scope of this exam, we use the bike sharing data set "bike". This dataset contains the hourly count of rental bikes between years 2011 and 2012 in Capital bikeshare system in Washington D.C. with the corresponding weather and seasonal information. Table 1 includes names and descriptions of the variables.

Table 1: Description of the data set

Variable	Description	Mean/Freq	Std. dev.	Min/max
cnt	hourly count of rental bikes	189.5	181.3876	1/977
season	1 = winter	4242		
	2 = spring	4409		
	3 = summer	4496		
	4 = fall	4232		
year (yr)	0 = 2011	8645		
	1 = 2012	8734		
daytime	night = 23:00 - 5:59	5004		
	morning = 6:00 - 9:59	2906		
	midday = 10:00 - 14:59	3640		
	afternoon = 15:00 - 18:59	2917		
	evening = 19:00 - 22:59	2912		
holiday	0 = yes	16879		
	1 = no	500		
temp	Normalized temperature in Celsius	0.4970	0.1926	0/1
hum	Normalized humidity	0.6272	0.1929	0/1
windspeed	Normalized wind speed	0.1901	0.1223	0/0.8507
weathersit	1 = Clear, Few clouds, Partly cloudy	11413		
	2 = Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist	4544		
	3 = Light Snow, Light Rain + Scattered clouds, Light Rain + Thunderstorm + Scattered clouds	1419		
	4 = Snow + Fog, Heavy Rain + Ice Pallets + Thunderstorm + Mist,	3		

Exercise 2: (22.5 Points)

Suppose you want to model the influence of some variables on the number of rented bikes per hour. Consider the following R Output:

```
# Call:
# lm(formula = cnt ~ season + yr + daytime + holiday + temp + hum +
#     windspeed, data = bike)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -361.34  -76.50  -16.14   55.26  557.62
#
# Coefficients:
#              Estimate Std. Error t value
# (Intercept)   -60.932     5.435  -11.210
# season2        37.108     3.302   11.239
# season3        18.977     4.255    4.460
# season4        66.390     2.861   23.205
# yr1            83.503     1.840   45.391
# daytimemorning 185.088     2.809   65.898
# daytimemidday 149.881     2.860   52.411
# daytimeafternoon 276.285     3.141   87.963
# daytimeevening 151.733     2.885   52.595
# holiday1      -27.722     5.466   -5.072
# temp          279.108     8.222   33.948
# hum          -137.247     5.471  -25.088
# windspeed     -42.244     7.938   -5.322
#
# Estimated model standard error: 120.3
# Multiple R-squared:  0.5604, Adjusted R-squared:  0.5601
```

a) How does the first column of the design matrix $\mathbf{X}$ look like in this specific example? Also name the dimensions. (1 Point)

b) What are the dimensions (i.e. number of rows and columns) of $(\mathbf{X}'\mathbf{X})^{-1}$ in this example? (1 Point)

- c) Carry out the test for significance of regression at the significance level $\alpha = 0.01$. Use the appropriate quantile of the correct distribution on page 9. (7 Points)
- i) Calculate the test statistic. (3 Points)
 - ii) Which quantile do you have to use (including distribution, degrees of freedom and critical value)? (2 Points)
 - iii) What is the decision of the test (with justification)? (1 Point)
 - iv) Interpret your result in the context of this exercise. (1 Point)

d) We want to carry out a composite test of a subvector with

$$\mathbf{C} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \text{ and } \mathbf{d} = \mathbf{0}_3$$

What are the specific null hypothesis and the alternative hypothesis here? (1 Point)

e) Carry out the test from part d) at the significance level $\alpha = 0.01$. Use the appropriate quantile of the correct distribution on page 9. (8 Points)

i) Suppose $\Delta SSE = 26731881$. Calculate the test statistic. (4 Points)

f) Calculate the 99%-confidence interval for β_{12} (**windspeed**). (3.5 Points)

g) Suppose $AIC = 215827.9$ and $BIC = 215936.6$ for the model in this exercise. What does this tell you about the model choice? (1 Point)

Exercise 3: (1.5 Points)

Now consider the following model:

```
# Call:
# lm(formula = cnt ~ season + yr + daytime + holiday + weathersit,
#     data = bike)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -341.10  -82.38  -13.67   61.87  549.38
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)    -79.494     2.723  -29.199  <2e-16
# season2         100.165     2.672   37.484  <2e-16
# season3         123.704     2.667   46.387  <2e-16
# season4          91.887     2.698   34.063  <2e-16
# yr1              91.658     1.884   48.647  <2e-16
# daytime morning  185.568     2.897   64.049  <2e-16
# daytime midday   194.170     2.705   71.769  <2e-16
# daytime afternoon 330.963     2.893  114.390  <2e-16
# daytime evening  177.227     2.895   61.228  <2e-16
# holiday1        -28.358     5.640   -5.028   5e-07
# weathersit2       -21.711     2.193   -9.901  <2e-16
# weathersit3       -89.733     3.503  -25.613  <2e-16
# weathersit4      -127.920    71.699   -1.784   0.0744
#
# Residual standard error: 124.1 on 17366 degrees of freedom
# Multiple R-squared:  0.5321, Adjusted R-squared:  0.5318
# F-statistic: 1646 on 12 and 17366 DF, p-value: < 2.2e-16
```

Explain why the p-value for `weathersit4` is comparatively high. Hint: Also Consider Table 1.

Quantile tables

Table 2: Quantiles for some t -distributions

quantile	0.495	0.9	0.95	0.975	0.99	0.995
3	-0.0136	1.6377	2.3534	3.1824	4.5407	5.8409
7	-0.0130	1.4149	1.8946	2.3646	2.9980	3.4995
12	-0.0128	1.3562	1.7823	2.1788	2.6810	3.0545
13	-0.0128	1.3502	1.7709	2.1604	2.6503	3.0123
50	-0.0126	1.2987	1.6759	2.0086	2.4033	2.6778
100	-0.0126	1.2901	1.6602	1.9840	2.3642	2.6259
17366	-0.0125	1.2816	1.6449	1.9601	2.3266	2.5761

Table 3: Quantiles for the standard normal distribution

quantile	0.495	0.9	0.95	0.975	0.99	0.995
	-0.0125	1.2816	1.6449	1.9600	2.3263	2.5758

Table 4: Quantiles for some F -distributions with fixed $df_2 = 17366$

quantile	0.495	0.9	0.95	0.975	0.99	0.995
$df_1 = 3$	0.7799	2.0841	2.6054	3.1169	3.7827	4.2808
4	0.8313	1.9452	2.3724	2.7866	3.3203	3.7164
5	0.8630	1.8476	2.2146	2.5672	3.0183	3.3512
6	0.8846	1.7744	2.0991	2.4090	2.8030	3.0926
7	0.9003	1.7171	2.0101	2.2883	2.6404	2.8981
8	0.9121	1.6705	1.9389	2.1925	2.5123	2.7456
9	0.9214	1.6319	1.8804	2.1144	2.4083	2.6223
10	0.9288	1.5991	1.8312	2.0491	2.3219	2.5201
11	0.9350	1.5708	1.7892	1.9935	2.2487	2.4337
12	0.9401	1.5462	1.7527	1.9455	2.1858	2.3595
13	0.9445	1.5244	1.7207	1.9035	2.1309	2.2950

Table 5: Quantiles for some χ^2 -distributions

quantile	0.495	0.9	0.95	0.975	0.99	0.995
$df = 3$	2.3395	6.2514	7.8147	9.3484	11.3449	12.8382
4	3.3249	7.7794	9.4877	11.1433	13.2767	14.8603
5	4.3151	9.2364	11.0705	12.8325	15.0863	16.7496
6	5.3077	10.6446	12.5916	14.4494	16.8119	18.5476
7	6.3017	12.0170	14.0671	16.0128	18.4753	20.2777
8	7.2966	13.3616	15.5073	17.5345	20.0902	21.9550
9	8.2921	14.6837	16.9190	19.0228	21.6660	23.5894
10	9.2881	15.9872	18.3070	20.4832	23.2093	25.1882
11	10.2844	17.2750	19.6751	21.9200	24.7250	26.7568
12	11.2810	18.5493	21.0261	23.3367	26.2170	28.2995
13	12.2779	19.8119	22.3620	24.7356	27.6882	29.8195
17366	17362.9977	17605.2608	17673.6763	17733.1597	17802.4894	17849.8010